GABRIEL CAVALCANTI ALVES

Full Stack Developer | Vue.js | Java | Spring Boot

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PROFESSIONAL SUMMARY

Full Stack Developer with hands-on experience in Vue.js, TypeScript, Java, and Spring Boot, gained while developing a comprehensive management system for the "Casa do Ancião" social project via the UFC Informatics Practice Center (NPI). I contributed to improving system data consistency and management, as well as front-end development. I have developed responsive interfaces, RESTful APIs, and automated tests using Playwright, while collaborating in Agile Scrum teams using Git/GitLab and CI/CD pipelines. I also possess a solid background in game development with Unity and C#, including published research on AI applied to games (SBGames 2025). Holds a degree in Software Engineering from UFC.

PROFESSIONAL EXPERIENCE

Informatics Practice Center (NPI) – UFC – Software Engineering Intern
October 2024 – August 2025 | Quixadá, CE

- Worked as a Full Stack Developer on the "Casa do Ancião" social project (a home for the elderly), developing a comprehensive staff management system using Vue.js, Vuetify, TypeScript, Java, Spring Boot, and PostgreSQL.
- Developed full-stack modules for staff management and other administrative processes for the institution.
- Built responsive interfaces, implemented form validations, and carried out CSS/UI improvements across multiple system modules.
- Built and maintained RESTful APIs using Java and Spring Boot, structuring requests and responses to ensure the integrity of CRUD operations for records.
- Conducted automated tests using Playwright, fixing critical bugs and HTTP 500 errors, thereby improving system stability.
- Identified and corrected discrepancies in digital records, resolving data integrity issues with the aid of DBeaver.
- Refactored code by improving data typing (changing from `varchar` to `TIME`) and replacing inline styles with Vuetify utility classes.
- Enhanced system performance through pagination, sorting features, and responsive tables.
- Collaborated within a Scrum team using GitLab, SonarQube, and CI/CD pipelines, while also using Figma to interpret UX/UI requirements.

PUBLISHED RESEARCH

"The Players' Perception Over The Artificial Intelligence in Digital Games: An Exploratory Study With Final Fantasy XV" — SBGames 2025.

GAME DEVELOPMENT PROJECTS (ADDITIONAL)

(Não) Foi o Ham! — 2.5D Investigation Game | Unity, C# - https://github.com/Muzamba/naoFoiOHam
August 2026 – Present | Team project (Lumo Lab)

- Collaborated on the development of a corporate-themed investigation game where the player is accused of stealing company files and must search for clues to prove their innocence and identify the real culprit.
- Implemented systems for dialogue boxes, clue interaction for game progression, and pausing.
- Developed a cross-scene persistence system for collected clues.
- Adapted the dialogue system to trigger during cutscenes.

Cook & Conquer — Survival Roguelike (Medieval/Cooking) | Unity, C# — github.com/PierrotNobre/Project_RED
March 2026 – Present | Team Project (TAPPS)

- Collaborated on the development of a medieval- and cooking-themed roguelike survival game, where the player navigates a dungeon, collecting ingredients and cooking recipes to power up before facing the final boss.
- Implemented combat systems against enemies with varied behaviors, an inventory system with item management, and enemy item drops.
- Developed a crafting system using cooking recipes based on Scriptable Objects, allowing for easy game balancing.
- Structured procedural tilemap generation to create dynamic, replayable dungeons, featuring a health system and progressive difficulty.
- Published multi-platform versions (WebGL and Android) with platform-specific optimizations.

Astro Adventure 3D – 3D Space Survival Game | Unity, C# — github.com/GabrielCAlves/Astro-Adventure-3D
August 2025 – Present

- Developed the 3D game *Astro Adventure* in Unity using C#, taking it from prototype to final build, including gameplay, combat systems, progression, and the user interface.

- Programmed character movement (walking, running, jumping, idle, and death) controlled by a State Machine, ensuring smooth transitions and organized code.
- Implemented a save/load system with checkpoints, persisting player progress, health, and items across sessions.
- Created a dual-weapon combat system and programmed basic enemy AI with health, damage, and death states, including a trigger-activated boss.
- Integrated Cinemachine for dynamic camera movement—including screen shake upon taking damage—and configured Post-Processing to enhance visual quality.
- Developed reusable C# components with modular scripts, accelerating the creation of new features.

Astro Obstacle — 3D Runner (Android) | Unity, C# — github.com/GabrielCAlves/Astro-Obstacle

August 2025 – Present

- Developed a 3D runner game for Android where the player navigates randomly generated platforms, dodging obstacles and collecting power-ups.
- Structured power-ups using inheritance to facilitate system expansion, including 3 distinct power-up types.
- Utilized DoTween for smooth animations, such as a bounce effect upon collecting power-ups and a scaling effect at game start.
- Implemented level generation with color randomization and a Level Manager featuring 3 level types.
- Optimized the game for mobile devices, including character animations and particle effects for power-up collection and death events.

Lost in The Stars – Action-Platformer 2D | Unity, C# — github.com/GabrielCAlves/Lost-In-The-Stars

May 2025 – Present

- Built systems for side-scrolling movement, jumping, and running with integrated animations (idle, run, jump); 2D combat (ranged shooting and contact damage); coin collection and counting; decoupled character health and death logic; and a camera system that follows the main character.
- Developed enemies with basic behaviors, implemented using code reuse.
- Managed scene transitions, the UI (coin counter and navigation buttons), and the audio system (ambient music, collection and jump sounds, and VFX such as particles for collecting, running, and jumping).

MyPong – Multiplayer Pong Game | Unity, C# — github.com/GabrielCAlves/MyPong

May 2025 – Present

- Modernized the classic game Pong with customizable colors and controls, as well as local multiplayer support.
- Designed an intuitive interface for quick match setup and a smooth gameplay experience.

4Kingdoms — 2D Fighting and Platformer Game (4 Players) | Unity, C# — github.com/GabrielCAlves/4-Kingdoms

February 2022 – July 2022 | University Project, UFC

- Developed a 2D fighting and platformer game for up to 4 players as a course project at UFC, set in a medieval world divided into four warring kingdoms.
- Implemented combat mechanics, movement, and a health system featuring healing potions and shields, distributed across three levels with distinct challenges.
- Structured support for four local players with independent controls and progression logic based on player elimination.

EDUCATION

Bachelor's Degree in Software Engineering | Federal University of Ceará (UFC) – Quixadá, CE

February 2020 – August 2025

CERTIFICATIONS

- Profession: Game Designer and Unity Developer — EBAC (2024–2025)
- Scrum Foundation Professional Certificate (SFPC) — CertiProf
- Cambridge English Level Certificate — Cultura Inglesa (2024)

TECHNICAL SKILLS

Frontend: Vue.js, TypeScript, JavaScript, HTML5, CSS3, Vuetify, Vite

Backend: Java, Spring Boot, Node.js, Strapi, REST APIs

Databases: PostgreSQL, MySQL, SQL, DBeaver, Firebase

Tools & DevOps: Git, GitLab, GitHub, CI/CD, SonarQube, Figma

Testing: Playwright, Testes Automatizados

Methodologies: Scrum, Agile, POO

Game Development (Complementary): Unity (2D and 3D), C#, Gameplay Programming, Level Design, Game AI, State Machines, Scriptable Objects, Cinemachine, DoTween, Post-Processing, VFX/Particles, Animation

LANGUAGES

- Portuguese — Native
- English — Intermediate / Advanced (B2+)